# Investigation of IGZO-Based 3D Vertical Channel Transistors and Physics-Informed Neural Network Modeling for Next-Generation DRAM

Wonbin Lee[1], Minje Park[1], and Saeroonter Oh[2]

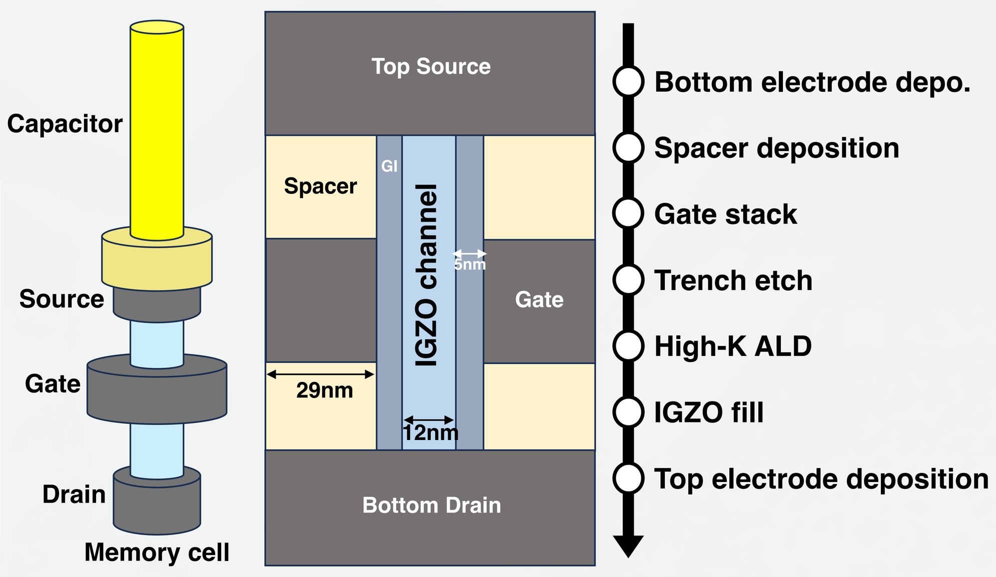
[1] Department of Electronic and Electrical Engineering, Sungkyunkwan University, Suwon, Republic of Korea
[2] Department of Semiconductor Convergence Engineering , Sungkyunkwan University, Suwon, Republic of Korea

## Research Motivation & Methods

- ✓ In conventional planar DRAM devices, higher leakage current lead to degraded retention
  → **IGZO-based VCT for future DRAM applications that meet key requirements** [1], [2]
- ✓ TCAD simulation: Analysis of short-channel effects, parasitic resistance and capacitance
  - Gate length
  - Channel doping
  - Asymmetric spacer thickness

**VCT requirements for $4F^2$ DRAM**

- On current > 1 $\mu$A/cell
- Low leakage < 1 fA/cell
- Small variability, SCE
- Small parasitic resistance/capacitance
- Stability against BEOL process in $H_2$ ambient



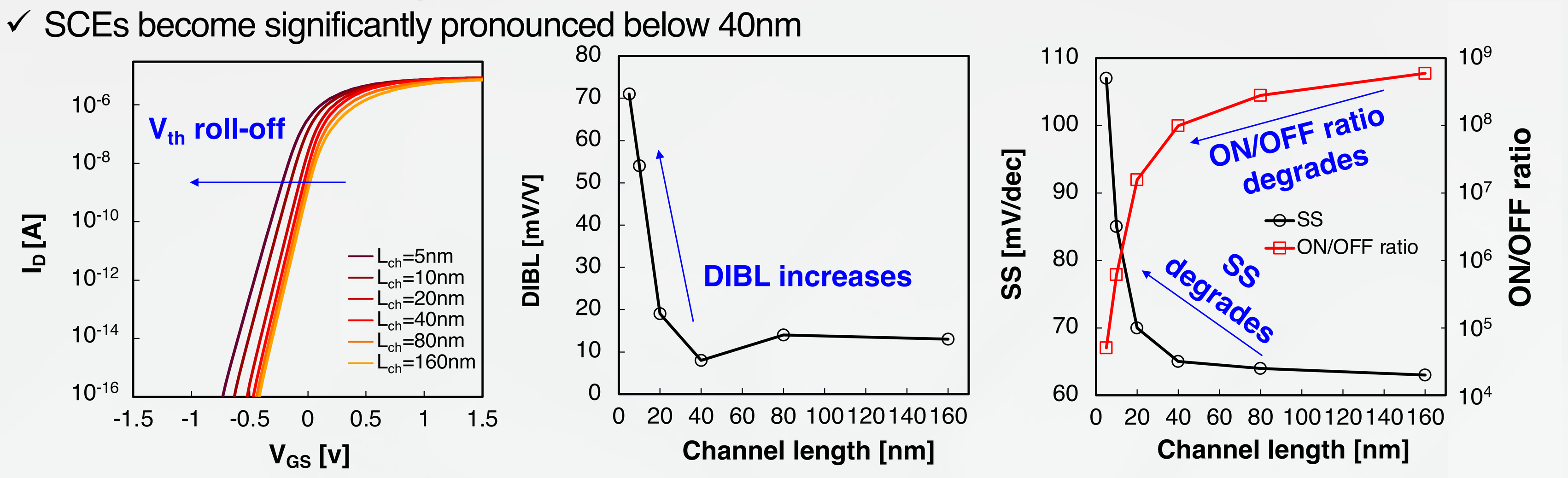

- ✓ When exploring new device concepts, simulating over the full parameter space are impractical
- ✓ To alleviate this bottleneck, we train a **Physics-Informed Neural Network (PINN) on 3D TCAD data** to predict device I–V characteristics efficiently while preserving key physical trends

➢ **This study demonstrates the feasibility of combining TCAD simulation and PINN to efficiently explore the device design parameter space to accelerate predictive analysis and data-driven design of IGZO-based VCTs for next-generation DRAM applications.**

## Device Parameter Space Exploration

### ❖ Device Scalability

✓ SCEs become significantly pronounced below 40nm

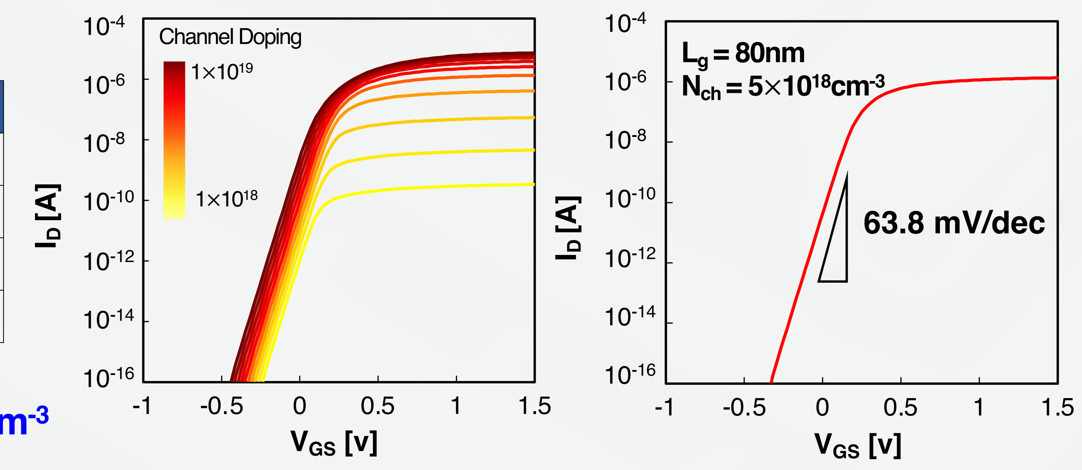



### ❖ DC Performance Optimization of VCT-DRAM ($L_G$, $N_{ch}$)

- ✓ WL voltage : 1.8V / -0.3V
- ✓ BL voltage : 1V / 0V

| | VCT-DRAM |
|---|---|
| $I_{D,on}$ | 1.39$\mu$A (@ 1.8V) |
| $I_{D,off}$ | 0.36fA (@ -0.3V) |
| SS | 63.8mV/dec |
| $V_{th}$ | 0.23V ($V_{th}$ extracted at 100nA) |

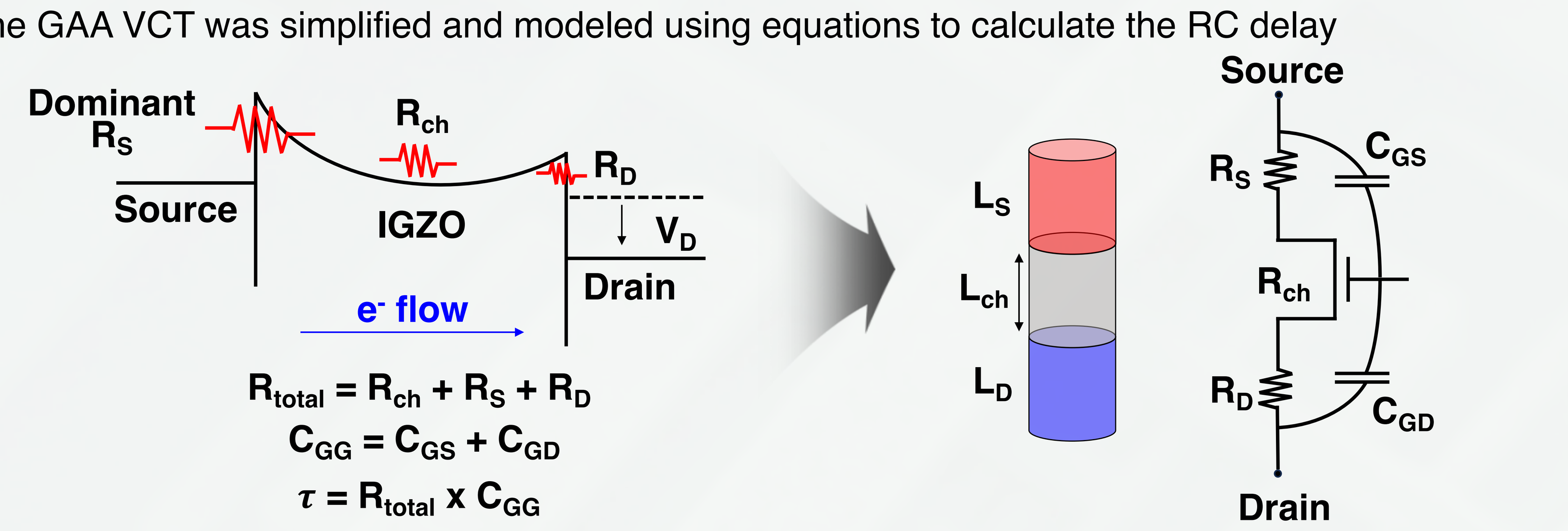
✓ **We identified optimal conditions**
**- Doping concentration : 5×10$^{18}$cm$^{-3}$**
**- Channel Length : 80nm**



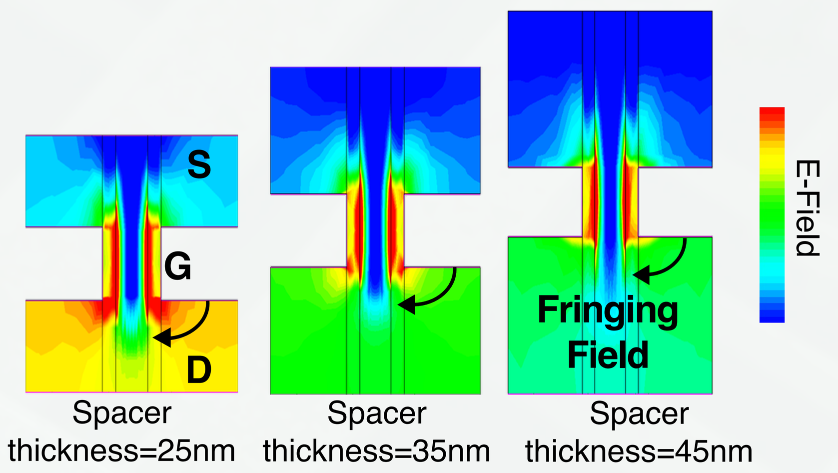

## Asymmetric Spacer Optimization Study

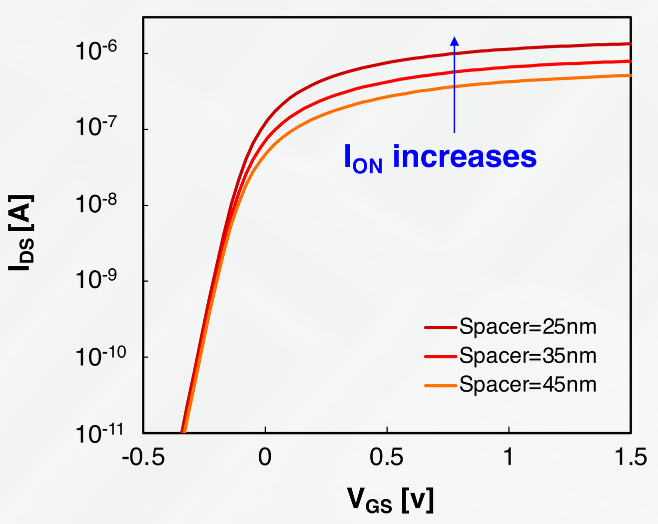
### ❖ Spacer Thickness Design

✓ The GAA VCT was simplified and modeled using equations to calculate the RC delay




- ✓ Parasitic R & fringing C mainly affect DC and AC performance
  → Reducing channel resistance enhances ON current
  → Better gate controllability by enhancing the fringing field

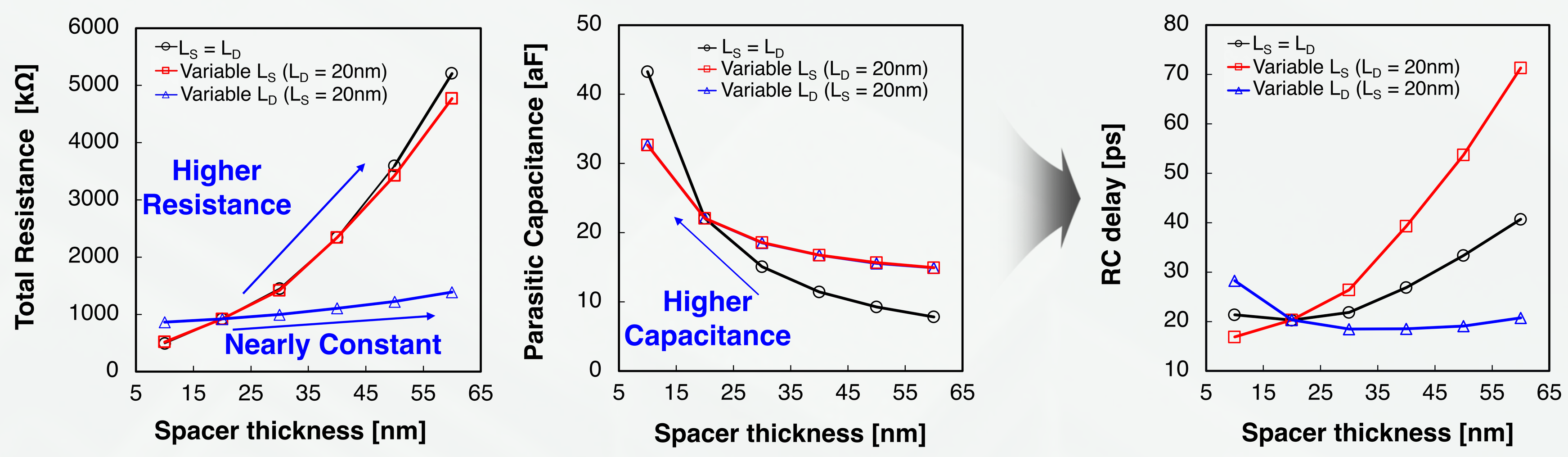


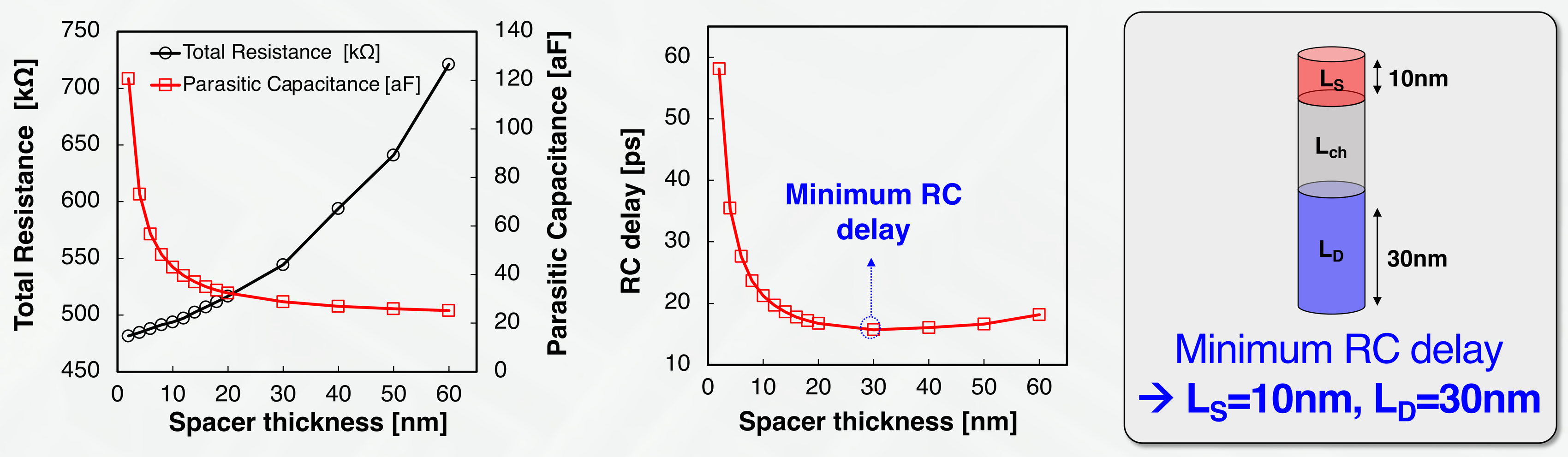

✓ **By thinning the spacer, On current was increased from 0.5$\mu$A to 1.3$\mu$A**

### ❖ AC Performance Optimization of VCT DRAM Structure

- ✓ Variable $L_D$ ($L_S$=20nm) shows nearly constant total resistance → **$L_S$ dominates overall device performance**
- ✓ **We found that the RC delay was minimized at $L_S$=10 nm and $L_D$=20nm**




✓ To further enhance performance , $L_D$ was varied while the $L_S$ was fixed at 10nm



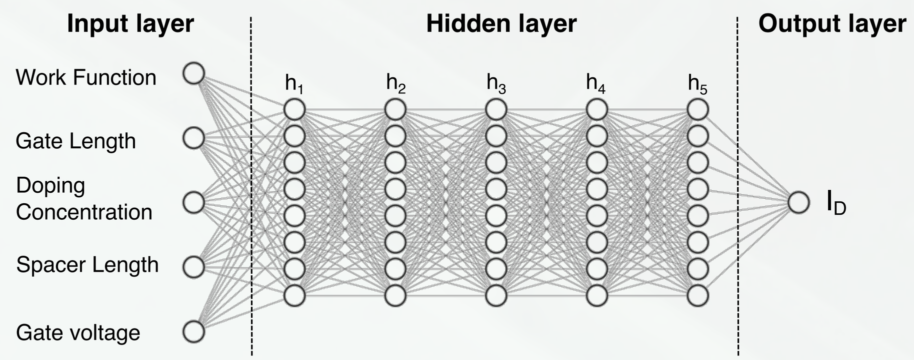

Minimum RC delay
**→ $L_S$=10nm, $L_D$=30nm**

## 3D TCAD-based Physics-Informed Neural Network Modeling

### ❖ I-V Modeling Using Physics Informed Neural Network

- ✓ Input data : Device Characteristic($d$) + $V_{GS}$
- ✓ Output data : $I_D$
- ✓ 5 hidden layer with skip connection

| Device Characteristic | Range |
|---|---|
| Work Function (eV) | 4.1 ~ 4.8 |
| Gate Length (nm) | 5, 10, 20, 40, 80, 160 |
| Doping Concentration (cm$^{-3}$) | 10$^{16}$ ~ 10$^{19}$ |
| Spacer Thickness (nm) | 15, 25, 35, 45 |



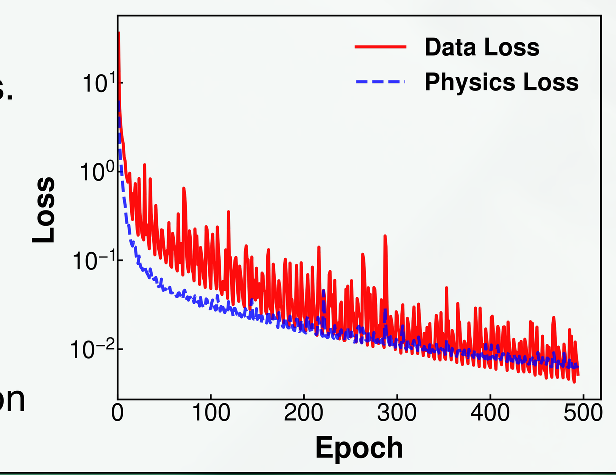

### ❖ Loss Function & Training Curve

- ✓ Mean Squared Loss + Physical Loss [3]
- ✓ Train the model to fit both the simulation data and I-V key trends.
- ✓ Training data is from 3D TCAD simulation of IGZO VCT

$$\mathcal{L} = \sum_{i=0}^{N} \|I_{data}(V) - I_{model}(d,V)\| + \lambda_1 \mathrm{E}\left[\frac{\partial^2 I}{\partial^2 V}\right] + \lambda_2 \mathrm{E}\left[\max\left\{0, -\frac{\partial I}{\partial V}\right\}\right]$$

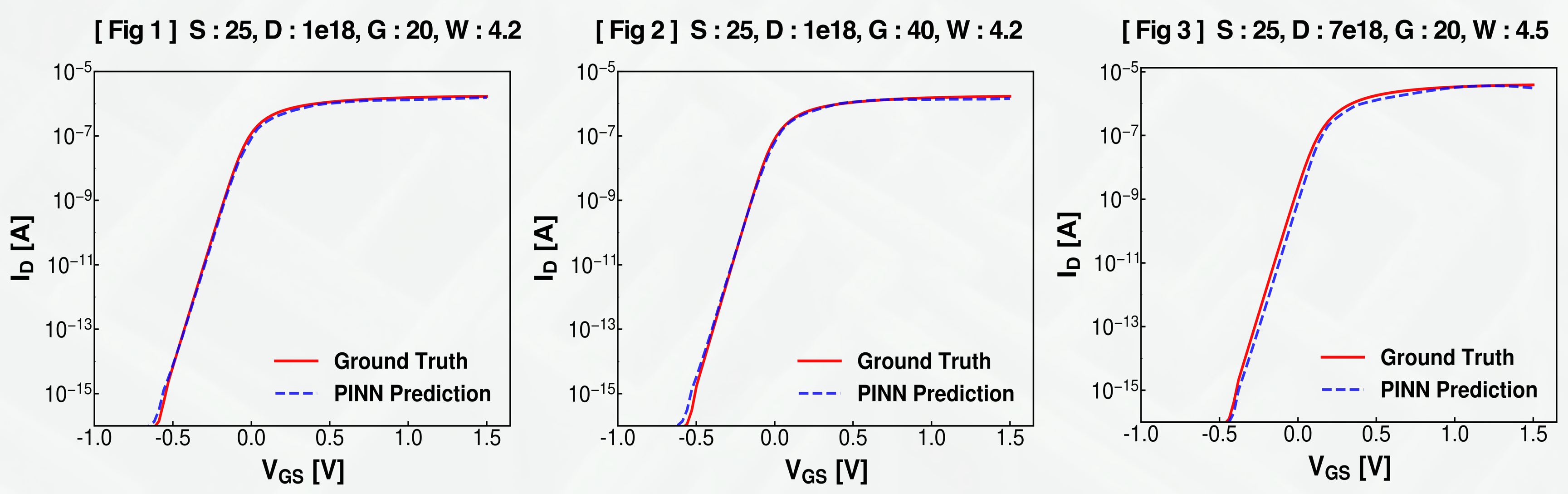
Smoothness Regularization : $\mathrm{E}\left[\frac{\partial^2 I}{\partial^2 V}\right]$  Monotonicity Loss : $\mathrm{E}\left[\max\{0, -\frac{\partial I}{\partial V}\}\right]$

→ In future work $\mathcal{L}$ will be adjusted to a physically based $I_D$ equation




### ❖ Model Performance

✓ Test device's characteristic hasn't been seen to the model during training session




| Prediction Target | Fig 1 | Fig 2 | Fig 3 |
|---|---|---|---|
| $I_{D,on}$ | 92.35% | 98.67% | 90.05% |
| $I_{D,off}$ | 90.54% | 81.95% | 82.92% |
| $V_{th}$ | 100.00% | 100.00% | 100.00% |

➢ Overall similarity across the test devices was **90.23% for $I_{D,on}$** (@1V), **83.81% for $I_{D,off}$** (@-0.3V), **and 95.17% for $V_{th}$**

## Conclusion

➢ To overcome the scaling limitations of conventional planar DRAM, IGZO-based VCT were optimized using TCAD. We identified the optimal conditions with a **doping concentration of 5x10$^{18}$cm$^{-3}$ and 80 nm channel length** satisfying the VCT requirements for $4F^2$ DRAM application.

➢ To further enhance device characteristics, an **asymmetric spacer configuration ($L_S$=10 nm, $L_D$=30 nm)** was implemented

➢ TCAD-generated dataset was used to train a PINN, where structural design parameters serve as inputs to predict the electrical device characteristics. Overall, similarity across the test devices were **90.23% for $I_{D,on}$** (@1V), **83.81% for $I_{D,off}$** (@-0.3V), **and 95.17% for $V_{th}$**

## References

[1] W. Zhao et al., "High-performance vertical gate-all-around oxide semiconductor transistors with 6 nm ALD IGZO channel and scaled contact CD down to 28 nm," *IEEE IEDM* 2024.
[2] S. Fujii et al., "Oxide-Semiconductor Channel Transistor DRAM (OCTRAM) with 4F² Architecture," *IEEE IEDM* 2024.
[3] M. Raissi et al., "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

## Acknowledgement

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